

Isadora Santos

AI & MLOps Engineer

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PROFESSIONAL SUMMARY

AI & MLOps Engineer with 5+ years of enterprise experience specializing in backend modernization, cloud infrastructure (AWS/Azure), and secure pipeline automation. Leveraging a strong SecOps background to build deployable AI applications and scale intelligent systems with rigorous software engineering practices.

TECHNICAL SKILLS

- **AI & Machine Learning:** Python, LangChain, LangGraph, LangSmith, ChromaDB, PostgreSQL pgvector, OpenAI API, Gemini API, Claude.
- **Engineering & MLOps:** Docker, Kubernetes (AKS/EKS), MLflow, Apache Airflow, FastAPI.
- **Backend & Databases:** Python, Java, SQL, JavaScript.
- **Cloud & Security:** AWS, Azure, SecOps, HashiCorp Vault, REST APIs, Microservices.

EDUCATION

- MLOps - Machine Learning Operations | Duke University | Present
- Artificial Intelligence Engineering Specialist Diploma | GRAN | Present
- Google AI Professional Certificate | In Progress
- Systems Analysis and Development Diploma | UNINASSAU
- Computer Programming | Georgian College
- Specialization in IT Management | Gran Concursos

FEATURED AI APPLICATIONS & MLOPS PROJECTS

Stateful SecOps Vulnerability Agent

- **Architected** an open-source LangGraph and Google Gemini API agent to automate triage and ticket generation, replicating a proprietary vulnerability management system built for Royal Bank of Canada (RBC).
- **Engineered** a containerized FastAPI and Docker backend with a ChromaDB Retrieval-Augmented Generation (RAG) system, utilizing PostgreSQL for reliable vector storage and GitHub Actions for automated deployment tasks.
- **Demo:** Watch a full video demonstration of this system in action at <https://isadorasantos.netlify.app/>

Secure MLOps Pipeline

- **Developed** a secure CI/CD MLOps pipeline mirroring RBC infrastructure to manage anomaly detection models, integrating continuous code quality checks (flake8) and automated security scanning (Trivy) via GitHub Actions.
- **Established** consistent model tracking and pipeline execution by containerizing the training environment and utilizing enterprise deployment standards like MLflow and LangSmith.
- **Demo:** Watch a full video demonstration of this system in action at <https://isadorasantos.netlify.app/>

Agentic Data Extraction & Self-Healing ETL

- **Built** an intelligent, LangGraph-driven ETL pipeline based on a system designed at RBC to dynamically map and extract complex, unstructured data into strict schemas.
- **Developed** an interactive React frontend for user validation, backed by a PostgreSQL data persistence layer to manage state and ensure reliable execution during complex workflows.

EXPERIENCE

Independent AI & DevSecOps Engineer | Montreal, QC | May 2025 - Present

- **Architected** production-grade AI apps (Stateful SecOps Agent, Agentic ETL) using Python, LangGraph, and the Google Gemini API, backed by secure PostgreSQL MLOps infrastructure and enterprise CI/CD practices (Trivy, MLflow, LangSmith).
- **Advanced** technical expertise through specialized programs in Machine Learning Operations (Duke University) and Artificial Intelligence (GRAN), applying rigorous theoretical frameworks directly to the development of enterprise-grade portfolio architectures.
- **Developed** *DDS na palma da mão*, an interactive WebAR workplace safety application using A-Frame and MindAR, and led university digital security extension initiatives focusing on scam prevention.

RBC - ROYAL BANK OF CANADA (Capital Markets) | Senior Engineer | Montreal, QC | Apr 2024 - Apr 2025

- **Architected** a Java Spring Boot and Azure OpenAI "AI Gateway" for 10+ teams, and led the legacy migration of monolithic settlement systems to Azure Kubernetes Service (AKS) microservices using Istio Service Mesh.
- **Designed** a secure Jenkins and MLflow continuous training pipeline for risk models enforcing strict Role-Based Access Control (RBAC).
- *Portfolio Replica:* Built the **Secure MLOps Pipeline** portfolio project to demonstrate these secure continuous training and deployment methodologies.

RBC - ROYAL BANK OF CANADA (Technology & Operations) | SecOps Engineer | Toronto, ON | Jan 2022 - Apr 2024

- **Built** an Apache Airflow, SQL, and AWS Lambda orchestration layer that automated enterprise vulnerability management, cutting Mean Time to Remediate (MTTR) by 60%.
- *Portfolio Replica:* Built the **Stateful SecOps Vulnerability Agent** portfolio project to demonstrate this automated threat triage and ticket generation architecture.
- **Developed** 'Self-Healing' Python infrastructure bots and an internal Cortex XSOAR cloud notification system, automatically correcting misconfigured cloud resources and reducing incident response times by 60%.

MUNICIPALITY OF YORK REGION | Statistical Data Analyst | Newmarket, ON | Jan 2021 - May 2021

- **Developed** automated Python data pipelines and secure ETL processes for sensitive health data (PHI), ensuring strict compliance with privacy regulations.
- *Portfolio Replica:* Built the **Agentic Data Extraction & Self-Healing ETL** portfolio project to demonstrate complex unstructured data mapping and reliable extraction workflows.
- **Engineered** a COVID-19 screening automation application using Microsoft PowerApps, SharePoint, and Power BI, managing the agile lifecycle and reducing manual errors by 70%.

LANGUAGES

- English: Fluent
- Portuguese: Fluent/Native
- Spanish: Advanced
- Russian: Intermediate
- French: Basic